Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 100 epochs with a batch size of 128. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the validation accuracy by dataset.

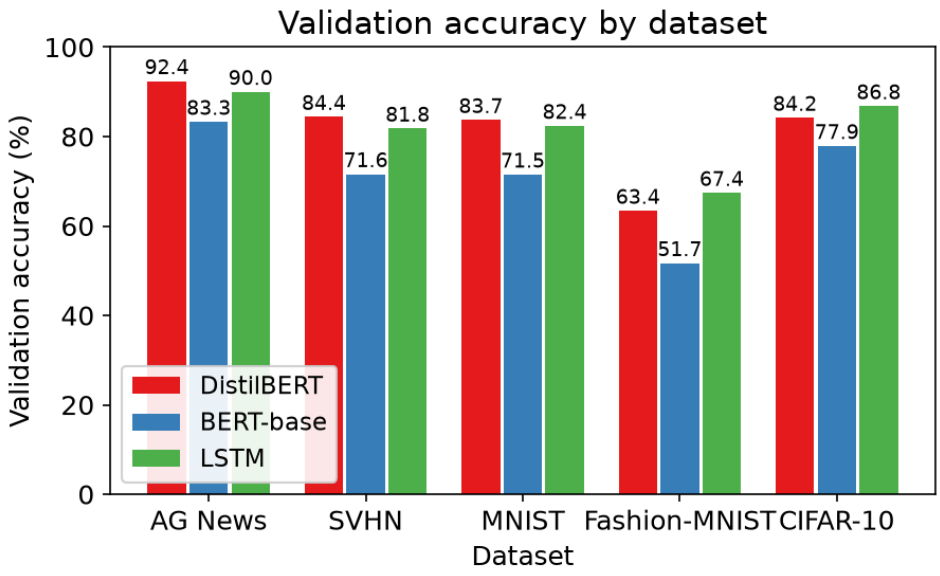


Figure 1: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 1, on Fashion-MNIST, LSTM obtains a validation accuracy of over 65%. DistilBERT outperforms LSTM by 2.4 percentage points on AG News. On SVHN, DistilBERT is 12.8 points better than BERT-base (Figure 1). DistilBERT achieves 42.4% validation accuracy on AG News (Figure 1).

Figure 2 shows the validation accuracy by dataset.

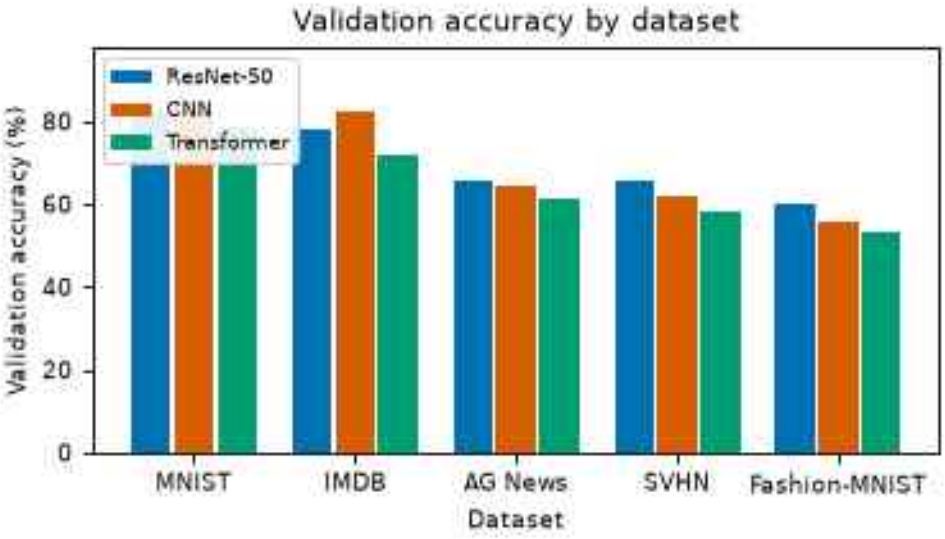


Figure 2: Validation accuracy by dataset for the models in this experiment.

On IMDB, CNN obtains a validation accuracy of 131.3%. As shown in Figure 2, CNN reaches 62.0% validation accuracy on SVHN.

Figure 3 shows the accuracy by dataset.

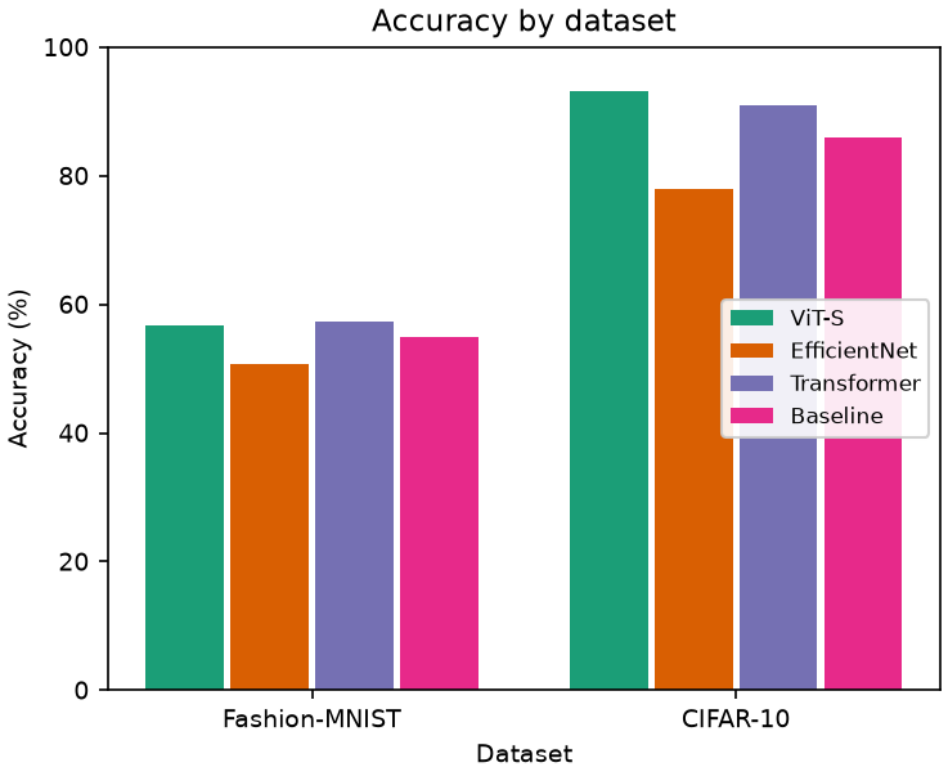


Figure 3: Accuracy by dataset for the models in this experiment.

EfficientNet reaches 55.0% accuracy on Fashion-MNIST (Figure 3). On CIFAR-10, Baseline obtains a accuracy of 78.0% (Figure 3).

Figure 4 shows the test accuracy by dataset.

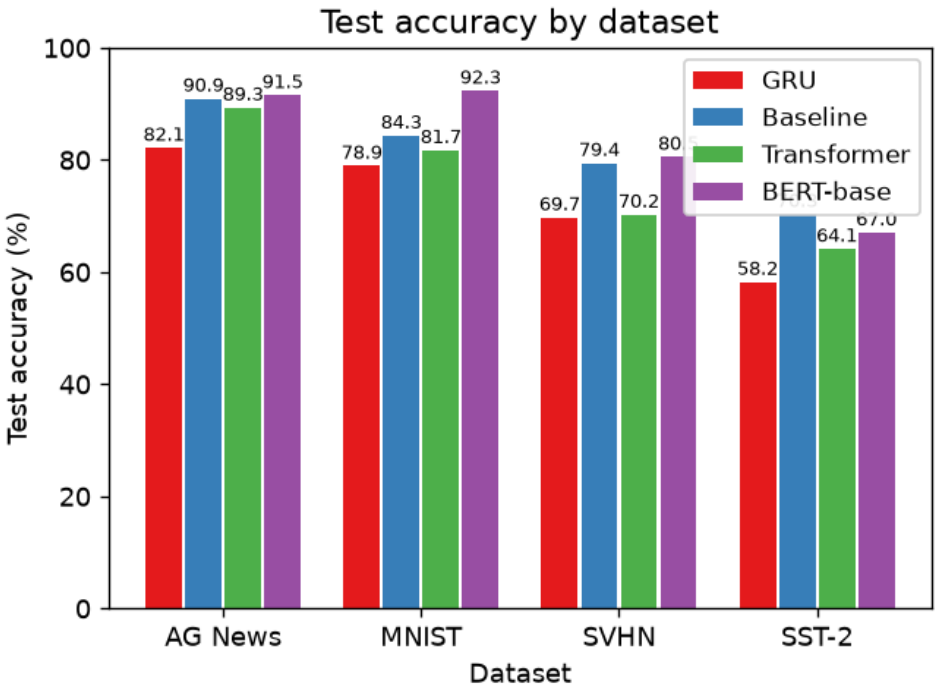


Figure 4: Test accuracy by dataset for the models in this experiment.

As shown in Figure 4, BERT-base outperforms Transformer by 15.6 percentage points on MNIST. Baseline achieves 70.3% test accuracy on SST-2.

3 Discussion

On SVHN, XGBoost obtains an accuracy of 52.6%. Overall, the results are in line with what we expected

from the literature.